

DIGITAL LOGIC DESIGN (CE_118)

Course overview

General Information

- Lecturer: Ho Ngoc Diem
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- **Course Assessment:**
 - Progress: 25%
 - Lab work: 25%
 - Final exam: 50%

Course Goals

- The primary goal is to provide in depth understanding of logic and system design, synthesis, and optimization for area, speed, and power consumption.
- Enable student to apply their knowledge for the design of advanced digital hardware system.
- Techniques for synthesizing FPGA/ASIC and the design of general-purpose processor.

Book & Material

1. Lecture notes
2. Principles of Digital Design, Daniel D. GAJSKI, Prentice Hall 1997
3. Digital Systems: Principles and Applications (eighth edition), Ronald J.Tocci.

Course Outline

- Chapter 1: Sequential Logic
- Chapter 2: Storage Components
- Chapter 3: Register-Transfer Design
- Chapter 4: Processor Design